

SQL Notes: ANY & ALL Operators

What is ANY?

ANY compares a value against a set of values.

The condition becomes TRUE if the comparison is true for at least one value in the set.

Think of it as:

"Does this comparison succeed for at least one element?"

```
SELECT *  
FROM Employees  
WHERE Salary > ANY  
(SELECT Salary  
FROM Managers  
);
```

```
WHERE ID = ANY  
(10, 20, 30  
)
```

Equivalent to

```
WHERE ID IN (10, 20, 30)
```

= ANY is rarely written in practice.

SHORTCUT

$x < \text{ANY}(\text{set})$



$x < \text{MAX}(\text{set})$

The optimizer often rewrites

`=ANY`

as

`IN`

For comparisons like

`>ANY`

MySQL may internally optimize using `MIN()`.

ALL is the opposite of ANY.

A comparison must be TRUE for every value in the set.

*Think
"Must satisfy all comparisons."*

Syntax

```
SELECT *  
FROM Employees  
WHERE Salary >ALL  
(SELECT Salary  
FROM Managers  
);
```

SHORTCUT

`x > ALL(set)`



`x > MAX(set)`

`x < ALL(set)`



`x < MIN(set)`

nstead of

```
WHERE Price <ALL  
(SELECT Price  
FROM Products  
)
```

people write

```
WHERE Price =  
(SELECT MIN(Price)  
FROM Products  
)
```